A Practical Refresher on Integration

Rules, Techniques, and Fully Worked Examples

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1 What Is Integration?

Integration is, in many cases, the reverse operation of differentiation.

If

$$F'(x) = f(x),$$

then F is an antiderivative of f, and

$$\boxed{\int f(x)\,dx = F(x) + C,}$$

where C is an arbitrary constant.

For example, since

$$\frac{d}{dx}(x^2) = 2x,$$

we have

$$\int 2x\,dx = x^2 + C.$$

For an *indefinite integral*, always include $+C$. Different constants have the same derivative, so integration cannot determine the constant.

2 Basic Integration Rules

2.1 Power Rule

For $n \neq -1$,

$$\boxed{\int x^n\,dx = \frac{x^{n+1}}{n+1} + C.}$$

Examples:

$$\int x^4\,dx = \frac{x^5}{5} + C,$$

and

$$\int \sqrt{x}\,dx = \int x^{1/2}\,dx = \frac{2}{3}x^{3/2} + C.$$

The case $n = -1$ is special:

$$\boxed{\int \frac{1}{x}\,dx = \ln|x| + C.}$$

2.2 Exponentials and Logarithms

$$\int e^x\,dx = e^x + C,$$

$$\int a^x\,dx = \frac{a^x}{\ln a} + C, \qquad a > 0, \quad a \neq 1,$$

and

$$\int \ln x\,dx = x\ln x - x + C, \qquad x > 0.$$

For a linear expression in the exponent,

$$\boxed{\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C, \quad a \neq 0.}$$

2.3 Trigonometric Integrals

$$\int \sin x\,dx = -\cos x + C,$$
$$\int \cos x\,dx = \sin x + C,$$
$$\int \sec^2 x\,dx = \tan x + C,$$
$$\int \frac{1}{1+x^2}\,dx = \arctan x + C,$$
$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C.$$

2.4 Linearity

Integration distributes over sums and constant multiples:

$$\int \big(af(x) + bg(x)\big)\,dx = a\int f(x)\,dx + b\int g(x)\,dx.$$

2.5 Worked Example: Integrate Term by Term

Evaluate

$$\int (4x^3 - 2x + 1)\,dx.$$

Step 1: Split the integral.

$$\int 4x^3\,dx - \int 2x\,dx + \int 1\,dx.$$

Step 2: Apply the power rule to each term.

$$4\frac{x^4}{4} - 2\frac{x^2}{2} + x + C.$$

Answer:

$$\boxed{x^4 - x^2 + x + C.}$$

3 Substitution (Change of Variables)

3.1 The Idea

Substitution is useful when an integrand contains a composite function and a factor related to the derivative of its inner function.

Set

$$u = g(x), \qquad du = g'(x)\,dx.$$

Rewrite the integral in terms of u, integrate, and substitute $g(x)$ back in.

3.2 Worked Example 1: Direct Substitution

Evaluate

$$\int 2x\cos(x^2)\,dx.$$

Step 1: Choose the inner function.

The expression x^2 is inside the cosine, and its derivative is $2x$. Set

$$u = x^2, \qquad du = 2x\,dx.$$

Step 2: Substitute.

$$\int 2x\cos(x^2)\,dx = \int \cos u\,du.$$

Step 3: Integrate and substitute back.

$$\int \cos u\,du = \sin u + C.$$

Since $u = x^2$,

$$\boxed{\int 2x\cos(x^2)\,dx = \sin(x^2) + C.}$$

Check:

$$\frac{d}{dx}\sin(x^2) = 2x\cos(x^2),$$

which matches the original integrand.

3.3 Worked Example 2: Substitution with an Exponential

Evaluate

$$\int xe^{x^2}\,dx.$$

Set

$$u = x^2, \qquad du = 2x\,dx.$$

Therefore,

$$x\,dx = \frac{1}{2}\,du.$$

Substitute:

$$\int xe^{x^2}\,dx = \frac{1}{2}\int e^u\,du.$$

Integrate:

$$\frac{1}{2}\int e^u\,du = \frac{1}{2}e^u + C.$$

Substitute $u = x^2$:

$$\boxed{\int xe^{x^2}\,dx = \frac{1}{2}e^{x^2} + C.}$$

3.4 Worked Example 3: A Linear Expression

Evaluate

$$\int \frac{3}{3x+2}\,dx.$$

Let

$$u = 3x + 2, \qquad du = 3\,dx.$$

Therefore,

$$\int \frac{3}{3x+2}\,dx = \int \frac{1}{u}\,du.$$

Using the logarithmic rule:

$$\int \frac{1}{u}\,du = \ln|u| + C.$$

Thus,

$$\boxed{\int \frac{3}{3x+2}\,dx = \ln|3x+2| + C.}$$

3.5 A Useful Pattern

If $g(x)$ is differentiable and $F' = f$, then

$$\boxed{\int f(g(x))g'(x)\,dx = F(g(x)) + C.}$$

In practice, look for an inner expression and a factor proportional to its derivative.

4 Integration by Parts

4.1 Formula and Intuition

Integration by parts comes from the product rule:

$$\frac{d}{dx}(uv) = u'v + uv'.$$

Rearranging and integrating gives

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

It is often useful for products where differentiating one factor simplifies it, such as a polynomial multiplied by an exponential, logarithm, or trigonometric function.

4.2 Worked Example 1: Polynomial Times Exponential

Evaluate

$$\int xe^x\,dx.$$

Step 1: Choose u and dv.
Choose

$$u = x, \qquad dv = e^x\,dx.$$

Then

$$du = dx, \qquad v = e^x.$$

Step 2: Apply the formula.

$$\int xe^x\,dx = xe^x - \int e^x\,dx.$$

Step 3: Finish the remaining integral.

$$\int xe^x\,dx = xe^x - e^x + C.$$

Therefore,

$$\boxed{\int xe^x\,dx = e^x(x-1) + C.}$$

4.3 Worked Example 2: Polynomial Times Logarithm

Evaluate

$$\int x \ln x \, dx, \qquad x > 0.$$

Choose

$$u = \ln x, \qquad dv = x \, dx.$$

Then

$$du = \frac{1}{x} \, dx, \qquad v = \frac{x^2}{2}.$$

Apply integration by parts:

$$\begin{aligned} \int x \ln x \, dx &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \frac{1}{x} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C. \end{aligned}$$

Hence,

$$\boxed{\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.}$$

4.4 How to Choose u

A useful heuristic is to choose as u the factor that becomes simpler when differentiated.

- $\ln x$: its derivative is $1/x$, often simpler.
- A polynomial such as x^3: repeated differentiation eventually gives zero.
- e^x or $\sin x$: these do not generally become simpler when differentiated.

This is a heuristic rather than an absolute rule; the best choice depends on the integral.

5 Partial Fraction Decomposition

5.1 When to Use It

Partial fractions are useful for rational functions, meaning quotients of polynomials, especially when the denominator factors into simpler expressions.

For example, a rational function may be rewritten as a sum of terms like

$$\frac{A}{x - a} + \frac{B}{x - b},$$

which are easy to integrate using logarithms.

5.2 Worked Example

Evaluate

$$\int \frac{1}{x^2-1}\,dx.$$

Step 1: Factor the denominator.

$$x^2-1=(x-1)(x+1).$$

Step 2: Decompose into partial fractions.
Write

$$\frac{1}{(x-1)(x+1)}=\frac{A}{x-1}+\frac{B}{x+1}.$$

Multiplying by $(x-1)(x+1)$ gives

$$1=A(x+1)+B(x-1).$$

Set $x=1$:

$$1=2A \quad\Longrightarrow\quad A=\frac{1}{2}.$$

Set $x=-1$:

$$1=-2B \quad\Longrightarrow\quad B=-\frac{1}{2}.$$

Therefore,

$$\frac{1}{x^2-1}=\frac{1}{2}\frac{1}{x-1}-\frac{1}{2}\frac{1}{x+1}.$$

Step 3: Integrate term by term.

$$\begin{aligned}\int \frac{1}{x^2-1}\,dx &= \frac{1}{2}\ln|x-1|-\frac{1}{2}\ln|x+1|+C\\ &= \boxed{\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right|+C}.\end{aligned}$$

The original integrand is undefined at $x=\pm 1$, so the antiderivative applies on intervals that do not cross those points.

6 Definite Integrals

6.1 The Fundamental Theorem of Calculus

If $F'(x)=f(x)$, then

$$\boxed{\int_a^b f(x)\,dx=F(b)-F(a).}$$

Unlike an indefinite integral, a definite integral has specified bounds and gives a number. Geometrically, it represents signed area: regions below the x-axis contribute negatively.

6.2 Worked Example

Evaluate

$$\int_0^2 3x^2\,dx.$$

An antiderivative is

$$F(x)=x^3.$$

Apply the bounds:

$$\begin{aligned}\int_0^2 3x^2\,dx &= \left[x^3\right]_0^2 \\ &= 2^3 - 0^3 \\ &= \boxed{8}.\end{aligned}$$

6.3 Definite Integrals and Substitution

Consider

$$\int_0^1 2x\cos(x^2)\,dx.$$

Let

$$u = x^2, \qquad du = 2x\,dx.$$

The bounds change as follows:

$$x = 0 \Rightarrow u = 0, \qquad x = 1 \Rightarrow u = 1.$$

Thus,

$$\begin{aligned}\int_0^1 2x\cos(x^2)\,dx &= \int_0^1 \cos u\,du \\ &= [\sin u]_0^1 \\ &= \boxed{\sin 1}.\end{aligned}$$

Alternatively, integrate first in x and then evaluate at the original bounds.

7 A Strategy for Unfamiliar Integrals

When you see an integral, try these steps in order:

1. **Simplify first.** Expand, factor, use exponent rules, or apply trigonometric identities when helpful.
2. **Try the basic rules.** Integrate powers and familiar functions term by term.
3. **Look for substitution.** Is there an inner function whose derivative also appears as a factor?
4. **Consider integration by parts.** Is it a product, with one factor becoming simpler when differentiated?
5. **For rational functions,** try polynomial division if needed, then partial fractions when the denominator factors.
6. **Check your answer.** Differentiate your proposed antiderivative. You should recover the original integrand.

Not every function has an antiderivative expressible using elementary functions. In such cases, numerical methods or special functions may be needed.

8 Practice Exercises

Try each exercise before reading the solutions. Remember to include $+C$ for indefinite integrals.

1. $$\int (5x^4 + 3x^2 - 2)\,dx$$

2. $$\int 6x(x^2+1)^4\,dx$$

3. $$\int x\cos x\,dx$$

4. $$\int \frac{1}{x^2-4}\,dx$$

5. $$\int_1^2 2x\,dx$$

Solutions

1. Basic Rules Integrate each term:

$$\begin{aligned}\int (5x^4 + 3x^2 - 2)\,dx &= 5\frac{x^5}{5} + 3\frac{x^3}{3} - 2x + C \\ &= \boxed{x^5 + x^3 - 2x + C}.\end{aligned}$$

2. Substitution Set

$$u = x^2 + 1, \qquad du = 2x\,dx.$$

Since $6x\,dx = 3\,du$,

$$\begin{aligned}\int 6x(x^2+1)^4\,dx &= 3\int u^4\,du \\ &= 3\frac{u^5}{5} + C \\ &= \boxed{\frac{3}{5}(x^2+1)^5 + C}.\end{aligned}$$

3. Integration by Parts Choose

$$u = x, \qquad dv = \cos x\,dx.$$

Then

$$du = dx, \qquad v = \sin x.$$

Apply the formula:

$$\begin{aligned}\int x\cos x\,dx &= x\sin x - \int \sin x\,dx \\ &= x\sin x + \cos x + C.\end{aligned}$$

Therefore,

$$\boxed{\int x\cos x\,dx = x\sin x + \cos x + C.}$$

4. Partial Fractions Factor:

$$x^2 - 4 = (x-2)(x+2).$$

The decomposition is

$$\frac{1}{x^2-4} = \frac{1}{4}\frac{1}{x-2} - \frac{1}{4}\frac{1}{x+2}.$$

Therefore,

$$\boxed{\int \frac{1}{x^2-4}\,dx = \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C.}$$

5. Definite Integral An antiderivative of $2x$ is x^2. Hence,

$$\begin{aligned}\int_1^2 2x\,dx &= \left[x^2\right]_1^2 \\ &= 4 - 1 \\ &= \boxed{3}.\end{aligned}$$

9 Final Checklist

Before moving on from an integral, ask:

- Did I choose the right rule or technique?
- Did I account for constants when substituting?
- Did I include $+C$ for an indefinite integral?
- For a definite integral, did I calculate $F(b) - F(a)$?
- Can I verify the result by differentiating?